Raritone AI-Powered Virtual Try-On Project

Challenges Faced and Solutions

Challenge 1: Limited Fashion Datasets

Public datasets may not contain enough diverse clothing styles, body types, and poses. Solution: Combined multiple datasets such as DeepFashion2, VITON-HD, and MPV.

Challenge 2: Human Pose Estimation Errors

Incorrect body landmark detection affects garment placement. Solution: Compared MediaPipe and OpenPose and selected MediaPipe for real-time performance.

Challenge 3: Clothing Segmentation Issues

Separating garments from complex backgrounds is difficult. Solution: Used U2Net and Detectron2 for accurate segmentation.

Challenge 4: Garment Alignment Problems

Clothes may not align correctly with different body shapes and poses. Solution: Implemented TPS and GMM warping techniques.

Challenge 5: Body Shape Variations

Different body sizes and postures affect fitting accuracy. Solution: Studied DensePose-based fitting approaches.

Challenge 6: Realistic Cloth Simulation

Simulating fabric folds and movement is challenging. Solution: Researched HR-VITON and advanced cloth warping models.

Challenge 7: Real-Time Processing Requirements

AI models can be computationally expensive. Solution: Used lightweight models and optimized workflows.

Challenge 8: Recommendation Accuracy

Basic recommendation methods may not reflect user preferences. Solution: Proposed Hybrid Recommendation Systems.

Challenge 9: Scalability Challenges

Supporting thousands of users requires significant resources. Solution: Suggested cloud-based deployment and GPU acceleration.

Challenge 10: AR/VR Integration Complexity

Developing immersive 3D try-on experiences requires advanced tools. Solution: Recommended phased implementation starting with 2D try-on.

Challenge 11: Mobile Device Compatibility

High-performance AI models may not run efficiently on mobile devices. Solution: Use model compression and cloud inference.

Challenge 12: User Privacy Concerns

User images and measurements require secure handling. Solution: Implement secure data storage and privacy policies.

Final Recommendation

Start with 2D Virtual Try-On, integrate Hybrid Recommendation Systems, add AI Size Prediction, expand to AR Live Try-On, and later develop 3D Avatar-Based Shopping.